

ELECTRIC POWER STEERING SYSTEM

High-Level Design Document

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Status: Approved for Engineering Analysis

Classification: Synthetic Demonstration Data

1. System Overview

The Electric Power Steering System coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

2. Software Architecture

The primary software components are SteeringManager, TorqueSensorControl, MotorDriveControl, AngleSensorManager, and DiagnosticManager.

3. Software Components

Component	Role	Primary Responsibility
SteeringManager	Coordinator	Computes assist torque from driver input.
TorqueSensorControl	Provider	Measures driver steering torque.
MotorDriveControl	Consumer	Drives the assist motor.
AngleSensorManager	Provider	Reports steering wheel angle.
DiagnosticManager	Consumer	Collects steering fault information.

4. Interfaces

Interface	Provider	Consumer	Signals
IDriverTorque	TorqueSensorControl	SteeringManager	DriverTorque
ISteeringAngle	AngleSensorManager	SteeringManager	WheelAngle
IAssistRequest	SteeringManager	MotorDriveControl	AssistCurrent
IFaultReport	SteeringManager	DiagnosticManager	FaultCode
IMotorFeedback	MotorDriveControl	SteeringManager	MotorSpeed

5. Ports

Component	Port	Direction	Interface
TorqueSensorControl	DriverTorque_Out	P-PORT	IDriverTorque
SteeringManager	DriverTorque_In	P-PORT	IDriverTorque
AngleSensorManager	SteeringAngle_Out	P-PORT	ISteeringAngle
SteeringManager	SteeringAngle_In	R-PORT	ISteeringAngle
SteeringManager	AssistRequest_Out	P-PORT	IAssistRequestLegacy
MotorDriveControl	AssistRequest_In	R-PORT	IAssistRequest
SteeringManager	FaultReport_Out	P-PORT	IFaultReport
DiagnosticManager	FaultReport_In	R-PORT	IFaultReport
MotorDriveControl	MotorFeedback_Out	P-PORT	IMotorFeedback
SteeringManager	MotorFeedback_In	R-PORT	IMotorFeedback

6. Signals

Signal	Data Type	Source	Destination
DriverTorque	int16	TorqueSensorControl	SteeringManager
WheelAngle	int16	AngleSensorManager	SteeringManager
AssistCurrent	int16	SteeringManager	MotorDriveControl
FaultCode	uint16	SteeringManager	TorqueSensorControl
MotorSpeed	uint16	MotorDriveControl	SteeringManager

7. Dependencies

Source	Relationship	Target	Reason
TorqueSensorControl	PROVIDES_INTERFACE	IDriverTorque	Publishes driver torque.
SteeringManager	REQUIRES_INTERFACE	IDriverTorque	Consumes driver torque.
AngleSensorManager	PROVIDES_INTERFACE	ISteeringAngle	Publishes wheel angle.
SteeringManager	PROVIDES_INTERFACE	IAssistRequest	Publishes assist current.
MotorDriveControl	REQUIRES_INTERFACE	IAssistRequest	Consumes assist current.
SteeringManager	PROVIDES_INTERFACE	IFaultReport	Publishes fault code.
DiagnosticManager	REQUIRES_INTERFACE	IFaultReport	Consumes fault code.
MotorDriveControl	PROVIDES_INTERFACE	IMotorFeedback	Publishes motor speed.
SteeringManager	REQUIRES_INTERFACE	IMotorFeedback	Consumes motor speed.

8. Functional Flows

Flow F-001: Driver Torque Distribution

TorqueSensorControl publishes DriverTorque through IDriverTorque. SteeringManager receives the signal through DriverTorque_In and updates its internal state model.

Flow F-002: Wheel Angle Distribution

AngleSensorManager publishes WheelAngle through ISteeringAngle. SteeringManager receives the signal through SteeringAngle_In and updates its internal state model.

Flow F-003: Assist Current Distribution

SteeringManager publishes AssistCurrent through IAssistRequest. MotorDriveControl receives the signal through AssistRequest_In and updates its internal state model.

Flow F-004: Fault Code Distribution

SteeringManager publishes FaultCode through IFaultReport. DiagnosticManager receives the signal through FaultReport_In and updates its internal state model.

Flow F-005: Motor Speed Distribution

MotorDriveControl publishes MotorSpeed through IMotorFeedback. SteeringManager receives the signal through MotorFeedback_In and updates its internal state model.

9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.